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NASA High Priority Open-Source Science (HPOSS) Program
Science Mission Directorate
National Aeronautics and Space Administration

Subject: Proposal Submission for the NASA HPOSS Program — TriadicFrameworks
Open-Science Infrastructure

Dear HPOSS Review Committee,

I am pleased to submit the enclosed proposal, "TriadicFrameworks: An Open-Science Substrate for Regime-Aware Modeling, Scanning, and Simulation," for consideration under the NASA High Priority Open-Source Science (HPOSS) program. This submission presents a unified, openly licensed scientific framework designed to support NASA's mission by enabling transparent, reproducible, and extensible analysis across sensing, modeling, and simulation environments.

TriadicFrameworks is an emerging open-science research initiative focused on developing substrate-agnostic protocols and tools for structural regime analysis. Recent publications, including the Dimensional Substrate Regime Scanning Protocol (dsrsp/0.1), available via Zenodo (DOI: 10.5281/zenodo.18331997), establish the foundation for a coherent, interoperable canon that supports scientific inquiry across planetary science, remote sensing, and autonomous exploration systems.

The proposed work aligns directly with HPOSS goals by:

- advancing open-source scientific infrastructure,
- providing community-accessible protocols and documentation,
- enabling reproducible workflows for sensor interpretation and regime classification, and
- supporting NASA's broader commitment to open science and transparent research practices.

Funding through HPOSS will support the development, documentation, and community integration of TriadicFrameworks components, including dsrsp/0.1, the Resonance Substrate Model (RSM), the Validation-Space-Time engine (vST), and associated open-source tooling. These deliverables will be released under permissive open licenses and maintained in public repositories to ensure long-term accessibility and scientific value.

Thank you for your consideration. I welcome the opportunity to contribute to NASA's open-science mission and to collaborate in advancing the state of open, reproducible scientific infrastructure.

Sincerely,

Nawder Loswin

Founder, TriadicFrameworks Research Initiative

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